BGIN IKP Working Group
Distinguishing Blockchain Forensics from Analytics

- BGIN is a multi-stakeholder discussion body / distributed think tank.

bg in

blockchain
governance
initiative
network

The Global Network for Blockchain Stakeholders™

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BGIN's Multi-stakeholder Approach

BGIN's distinctive multi-stakeholder approach provides a uniquely valuable foundation for advancing blockchain governance research. As a neutral facilitator with global reach and technical credibility, BGIN orchestrates collaborative processes that balance diverse interests across jurisdictional boundaries while maintaining focus on public good outcomes. This approach enables candid discussion of sensitive methodological issues that might be challenging in more commercially oriented environments. The organization's commitment to transparent development with broad stakeholder input, combined with its educational focus that translates complex technical concepts for varied audiences, creates an inclusive ecosystem where best practices can emerge organically. Through regular workshops, discussion forums, and public consultations, BGIN harnesses collective expertise spanning technical, legal, and regulatory domains to develop frameworks that are both technically sound and practically applicable across diverse global contexts. This collaborative methodology ensures that emerging standards benefit from comprehensive perspectives while remaining adaptable to evolving technological and regulatory landscapes.

Distinguishing Blockchain Forensics from Analytics

BGIN Identity, Key Management & Privacy Working Group 2025 Working References:

BGIN IKP Working Group Distinguishi…	Forensics v Analysis 30th Jan.pptx
IKP-WG Meeting Agenda (2025-01-30)	Bitcoin Fog Case Summary.pptx
Website_Tentative_Program_BGIN_B…	Bitcoin Fog Case Daubert Hearing Su…
	IKP_Forensics or Analysis

Executive Summary

Recent developments in blockchain investigation methodologies have highlighted the critical distinction between blockchain analytics and forensics. This distinction became particularly evident in the Bitcoin Fog case, which raised important questions about evidentiary standards, methodological rigor, and the legal admissibility of blockchain analysis.

The BGIN IKP Working Group is developing a framework to address these challenges through a multi-stakeholder approach, bringing together analytics companies, law enforcement, exchanges, legal experts, academics, and regulators to establish clear standards and best practices.

This document outlines our research approach, key findings to date, and next steps toward creating a comprehensive framework that can serve the global blockchain ecosystem.

1. Research Context

1.1 Background

The blockchain analytics industry has evolved rapidly, with companies developing sophisticated tools to analyse on-chain activity. These tools serve various purposes, from compliance monitoring and risk management to criminal investigation and legal proceedings. However, a critical distinction has emerged between:

- **Blockchain Analytics**: Probabilistic analysis of on-chain data to identify patterns and make claims about activity and actors. Analytics typically employs statistical methods, machine learning, and pattern recognition to draw inferences about blockchain activities without necessarily establishing definitive connections to real-world identities. These tools are primarily designed for risk management, compliance, and investigative leads.

- **Blockchain Forensics**: Meticulous, verifiable work of tracing transactions and establishing confirmed connections between individuals and public keys for evidentiary purposes. Forensic analysis must meet higher standards of proof, with methodologies that can withstand legal scrutiny and challenge. It involves rigorous documentation, chain of custody preservation, and verification protocols beyond pattern recognition.

The Bitcoin Fog case highlighted tensions between these approaches when blockchain analysis was presented as forensic evidence in court. Roman Sterlingov was charged with money laundering related to his alleged operation of a bitcoin mixer called Bitcoin Fog. The prosecution relied heavily on Chainalysis Reactor software to establish connections between Sterlingov and the service. The court's acceptance of these findings as admissible evidence, despite defence challenges from CipherTrace experts, established an important precedent that merits careful analysis.

1.2 Key Research Questions

1. **Evidentiary Thresholds**: When does pattern analysis become admissible evidence? What level of certainty is required to transform analytics into forensically sound evidence?
2. **Methodological Standards**: What scientific and technical standards should govern blockchain forensic tools and methodologies? How can these standards be consistently applied across different jurisdictions?
3. **Proprietary Technology Evaluation**: How should proprietary tools and methodologies be evaluated in legal proceedings? What level of transparency is necessary while respecting intellectual property concerns?
4. **Chain of Custody**: What constitutes an adequate chain of custody in blockchain investigations? How should on-chain data be preserved, verified, and presented as evidence?
5. **Commercial vs. Transparency Balance**: How can we balance the commercial interests of analytics companies with transparency requirements in forensic contexts? What disclosure standards are appropriate?
6. **Practitioner Qualifications**: What expertise should be required of individuals testifying about blockchain analysis? How should their qualifications be evaluated?
7. **Cross-Jurisdictional Standards**: How do evidentiary requirements vary across legal systems, and what principles can be universally applied?

2. The Analytics-Forensics Distinction

2.1 Fundamental Differences

Blockchain Analytics	Blockchain Forensics

Makes probabilistic claims based on patterns and heuristics	Requires verifiable connections between individuals and keys with demonstrable evidence
May rely on proprietary "black box" methodologies with limited external validation	Must meet rigorous evidentiary standards (e.g., Daubert in US, similar standards internationally)
Focused on risk scoring, pattern identification, and compliance monitoring	Focused on establishing facts with sufficient certainty for legal proceedings
Prevention of tx's rather than post	Post tx disclosures of real world details
Can incorporate behavioural and statistical models with varying levels of transparency	Requires methodological transparency, scientific rigour, and reproducibility
Commercial product prioritising efficiency, speed, and broad coverage	Legal evidence requiring verification, scrutiny, and documented methodology
Often incorporates machine learning and AI with evolving models	Typically requires stable, explainable methodologies with known error rates
Primarily serves compliance, risk management, and investigative purposes	Serves asa foundation for prosecution, asset seizure, and criminal proceedings

'To the weight not admissibility'

- We are not saying these are an either / or situation.

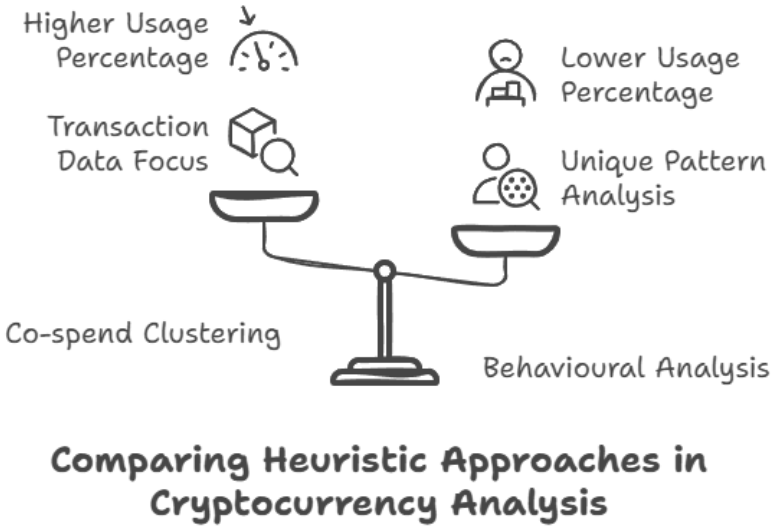
WHY IT MATTERS

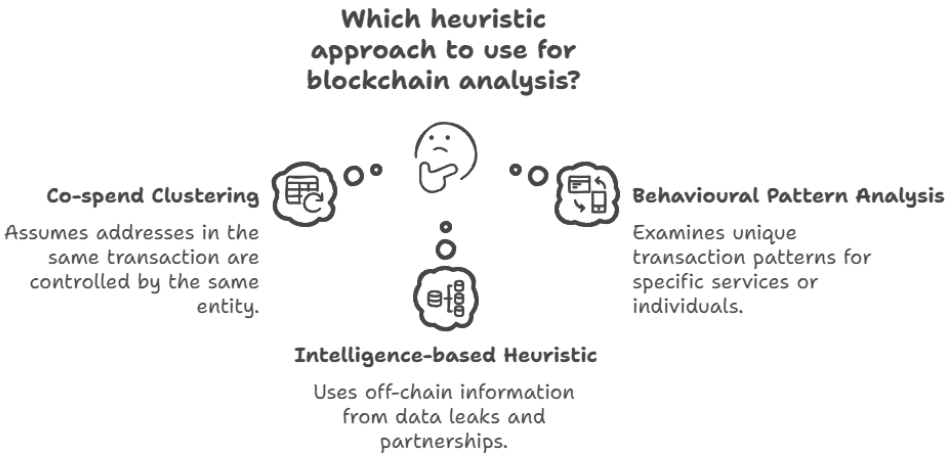
2.2 The Bitcoin Fog Case: Key Learnings

The Bitcoin Fog case provided important insights into how courts evaluate blockchain analysis tools:

1. **Heuristic Approaches**: Chainalysis Reactor relied on two primary heuristics:
    - *Heuristic 1 (50.26%): Co-spend/common spend clustering, which assumes addresses used as inputs in the same transaction are controlled by the same entity*

- *Heuristic 2 (49.74%): Entity-specific behavioural pattern analysis, examining transaction patterns unique to particular services or individuals*
- *Additionally, Heuristic 3 (not used in this case): Intelligence-based heuristic using off-chain information from sources like data leaks, court documents, and exchange partnerships*





2. **Defence Challenges**: CipherTrace's expert report raised several significant concerns:
    - Lack of independent verification and peer review of Chainalysis data and methodologies
    - Vague and potentially over-inclusive methodologies, particularly regarding Heuristic 2
    - Potential for false negatives that could falsely link innocent addresses to illicit activities
    - Significant discrepancies between Chainalysis attributions and CipherTrace findings

- Limited transparency in how specific sub-categories of Heuristic 2 were applied

3. **Court's Ruling**: Judge Randolph Moss determined that:
    - Blockchain analytics meets reliability standards under Rule 702 and Daubert criteria
    - Co-spend heuristics have received sufficient academic approval, while Heuristic 2 variability was deemed acceptable given its case-specific nature
    - The lack of a compiled "error rate" does not undermine reliability, based on testimony about the tool's accuracy in practice
    - Widespread use by law enforcement globally, financial institutions, and regulators indicates acceptance within the relevant community
    - The corroboration of findings through manual tracing and other software tools strengthened reliability
4. **Implications**: The ruling demonstrates:
    - Courts may accept proprietary methodologies without complete transparency if other reliability factors are present
    - Corroboration through multiple tools and manual verification significantly strengthens admissibility
    - Technical fields may be held to different peer review standards than traditional sciences
    - Practical experience and operational validation may substitute for formal error rate analysis
    - The court's gatekeeper role focuses on methodology reliability, not necessarily conclusion accuracy
5. **Post-Case Industry Response**:
    - Chainalysis publicly celebrated the ruling as validation of its methodologies
    - Emphasised the importance of having experts who can meticulously detail attribution processes
    - Warned customers against providers who claim visualisations alone are sufficient
    - Highlighted the need for comprehensive expert witness preparation

Data Analysis vs Forensic Analysis

- Each clustering technique combines multiple data types depending on the purpose or case.
- The use of techniques tailored to non-universal human behaviors, product characteristics, or specific cases, especially when utilizing data not publicly accessible, can be defined as forensic analysis.

Clustering Techniques	Data Type (1) on-chain data	Data Type (2) (2) Data identified through data leaks and undercover investigation	Data Type (3) (3) Data identified through governmental subpoena power	Data Type (4) (4) Data shared by entities
Clustering based on technical and scientific theories (e.g., co-spend clustering, multi-input clustering)	This is the primary data type used. Transaction histories and relationships between addresses are analyzed.	May be used as supplemental information (e.g., for verification or to provide additional context for clusters)	Typically not used, but legally obtained data might contribute to improving clustering accuracy in certain cases	Can be used when provided
Clustering based on the behavioral patterns of individuals or organizations and the characteristics of products.	Used to analyze how specific groups of addresses are being analyzed	May help identify behavioral patterns	Can be used to supplement details about behavioral patterns	Used when behavioral data is directly provided by clients or services
Clustering based on manual processes tailored to specific cases using data such as attribution	May not always be the primary data type but can be used depending on the case	Critical in specific case-based manual processes	Used to support manual approaches in specific cases	Frequently used for custom solutions tailored to specific cases

source: https://storage.courtlistener.com/recap/gov.uscourts.dcd.232431/gov.uscourts.dcd.232431.159.1.pdf
https://caselaw.findlaw.com/court/us-dis-crt-dis-col/115886148.html

Data analysis vs Forensic analysis

3. Methodological Frameworks

3.1 Clustering Techniques and Data Types

Clustering Technique	Primary Data Type	Secondary Data Types	Forensic Value	Academic Recognition	Challenges
Technical/Scientific Clustering (e.g., co-spend)	On-chain transaction data	May incorporate exchange data for verification	High - based on cryptographic principles	Strong - published in peer-reviewed literature	Limited by mixing services and CoinJoin techniques
Behavioural Pattern Clustering	On-chain activity patterns	Off-chain intelligence, service characteristics	Medium - depends on pattern specificity	Limited - often proprietary methodologies	Potential for false positives based on similar behaviours
Entity-Specific Clustering	Known entity behaviour	Data leaks, exchange information,and manual research	Variable - depends on data quality	Minimal - highly specific to individual cases	Difficult to generalise or independently verify
Manual Attribution	Multiple sources including human intelligence	Undercover operations, informants, subpoena results	High - when properly documented	Case-dependent	Resource-intensive, difficult to scale

Each technique combines multiple data types and methodologies:

1. **On-chain Data**: Public blockchain transaction data accessible to all participants
2. **Data from Leaks/Investigations**: Information obtained through data breaches or undercover operations
3. **Subpoena-Derived Data**: Information obtained through legal compulsion from exchanges and services
4. **Voluntarily Shared Data**: Information provided by exchanges, services, or other entities

The forensic value of these techniques varies based on:

- Reproducibility by independent parties
- Transparency of methodology
- Scientific basis and validation
- Chain of custody documentation
- Error rate calculation and disclosure

3.2 Key Concerns for Forensic Applications

1. **Data Access and Transparency**:
    - The proprietary nature of some data sources creates "black box" concerns
    - Limited ability to verify attribution sources independently
    - Commercial interests in protecting methodologies conflict with legal scrutiny requirements
    - Challenges in reproducing results without access to identical data sets
2. **Over-inclusivity Risk**:
    - Potential for false positives affects individuals' rights
    - Behavioural heuristics may capture innocent activities that resemble suspicious patterns
    - Risk of confirmation bias in manual attribution
    - Inadequate controls for distinguishing similar but unrelated activity patterns
3. **Error Rate Documentation**:
    - Lack of standardised error measurement methodologies
    - Insufficient benchmarking across different analysis platforms
    - Unclear procedures for calculating false-positive/negative rates
    - Challenges in creating test data sets for validation
4. **Market Concentration**:

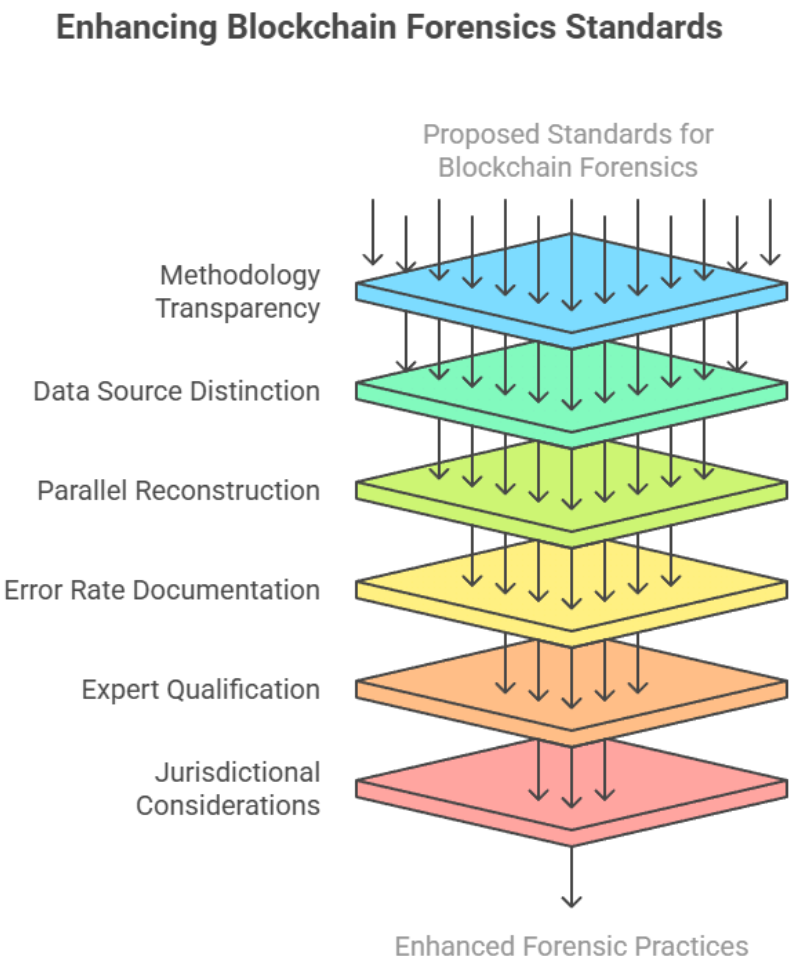
    - Risk of only established firms meeting "widespread acceptance" criteria
    - Barriers to entry for new forensic tool providers
    - Limited incentives for transparency in dominant market players
    - Potential conflicts when the same firm provides both analytics and forensic services
5. **Cross-Jurisdictional Inconsistency**:
    - Varying standards for admissibility across legal systems
    - Inconsistent application of scientific evidence standards
    - Different requirements for expert witness qualification
    - Challenges in international investigation coordination
6. **Methodological Evolution**:
    - Rapid development of blockchain technologies outpacing forensic standards
    - Emergence of privacy-enhancing technologies challenges existing techniques
    - Increasing sophistication of obfuscation methods
    - Need for continuous validation of methodologies against evolving threats

4. Toward a Best Practices Framework

4.1 Proposed Standards for Blockchain Forensics

1. **Methodology Transparency**:
    - Clear documentation of clustering techniques and their scientific basis
    - Explicit disclosure of limitations and potential failure scenarios
    - Distinction between established heuristics and experimental approaches
    - Documentation of version control and methodological changes over time
    - Peer-reviewable description of general methodologies while protecting specific implementations
2. **Data Source Distinction**:
    - Explicit identification of on-chain versus off-chain data sources
    - Classification system for data reliability and verification levels
    - Clear documentation of the chain of custody for all data sources
    - Procedures for handling potentially compromised data
    - Standards for data collection, storage, and preservation
3. **Parallel Reconstruction**:
    - Independent verification of key findings through alternative methods
    - Documentation of verification procedures and results
    - Standards for acceptable verification methodologies
    - Protocols for resolving discrepancies between approaches
    - Requirements for manual verification of automated findings
4. **Error Rate Documentation**:
    - Standardised approach to false-positive/negative reporting
    - Regular validation against known-outcome test cases
    - Independent testing and validation procedures
    - Statistical significance standards for findings
    - Confidence intervals for probabilistic claims
5. **Expert Qualification**:
    - Clear standards for expert witness qualification in blockchain forensics

    - Required training and certification frameworks
    - Continuing education requirements for practitioners
    - Ethical standards and conflict of interest disclosures
    - The distinction between tool providers and forensic analysts using the tools
6. **Jurisdictional Considerations**:
    - Adaptable framework for different legal systems
    - Jurisdiction-specific implementation guidelines
    - International standards for cross-border investigations
    - Procedures for managing jurisdictional conflicts
    - Recognition of varying evidentiary thresholds across systems
7. **Tool Validation Requirements**:
    - Independent testing and validation protocols
    - Regular auditing and review procedures
    - Change management documentation requirements
    - Version control and archival procedures
    - Transparency in algorithmic decision-making
8. **Documentation Standards**:
    - Comprehensive case documentation requirements
    - Standardised reporting formats and terminology
    - Chain of custody documentation procedures
    - Evidence preservation protocols
    - Disclosure requirements for methodological limitations



4.2 Areas Requiring Further Research

1. **Cross-Jurisdictional Analysis**:
    - Expansion beyond US-centric cases to include EU, Asian, and other legal systems
    - Comparative study of evidentiary standards across jurisdictions
    - Analysis of successful cross-border investigations and prosecutions
    - Documentation of jurisdictional conflicts and resolutions
    - Development of harmonised international standards
2. **TradFi Precedents**:
    - Study of relevant traditional finance evidentiary standards and their applicability
    - Analysis of financial forensics methodologies in banking and securities investigations
    - Lessons from anti-money laundering investigation standards
    - Application of existing financial evidence frameworks to blockchain
    - Adaptation of established forensic accounting principles
3. **Practitioner Certification**:
    - Exploration of qualification frameworks for blockchain investigators
    - Development of standardized training curricula
    - Certification standards and testing methodologies
    - Continuing education requirements
    - Ethical guidelines and professional standards
4. **Technical Standards Development**:
    - Establishment of baseline methodological requirements
    - Development of reference implementations for common techniques
    - Creation of test datasets for validation purposes
    - Standardisation of terminology and methodology descriptions
    - Implementation guidelines for blockchain forensic tools
5. **Data Hygiene Protocols**:
    - Definition of standards for attribution data quality
    - Procedures for data validation and verification
    - Methods for identifying and correcting data errors
    - Standards for data source reliability assessment
    - Protocols for handling conflicting data from multiple sources
6. **Emerging Technology Adaptation**:
    - Analysis of privacy coin investigation methodologies
    - Approaches to layer-2 solution forensics
    - Standards for cross-chain investigation
    - Adaptation to zero-knowledge proof systems
    - Forensic approaches to DeFi protocols and DAOs
7. **Stakeholder Impact Assessment**:
    - Analysis of blockchain forensics' impact on various stakeholders
    - Development of fairness and equity standards
    - Privacy impact assessment frameworks
    - Procedures for contesting false attributions
    - Remediation standards for incorrect attributions

Discussion - >> Conclusion

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6. Next Steps

6.1 Actions

1. **Educational Resource Development**:
    - Create briefing materials on blockchain analysis fundamentals
    - Develop contextual information packages for stakeholder interviews
    - Produce fact files for structured discussions on key topics
    - Draft technical glossary and terminology standards
    - Prepare workshop materials for BGIN Block meeting
2. **Expert Interview Program**:
    - Develop a structured interview protocol covering key research questions
    - Identify and recruit experts from diverse stakeholder groups
    - Conduct interviews with analytics companies, law enforcement, legal experts, and academics
    - Document and analyse interview findings
    - Synthesise insights into draft framework recommendations
3. **Case Study Development**:
    - Expand analysis beyond Bitcoin Fog to include diverse jurisdictions
    - Document methodologies used in successful prosecutions
    - Analyse defence challenges and court responses
    - Identify best practices from exemplary cases
    - Create anonymised templates for procedural documentation
4. **Technical Framework Development**: (LEXICON)
    - Establish baseline standards for forensic methodologies
    - Define validation requirements for common techniques
    - Create reference implementations for key heuristics
    - Develop error rate calculation methodologies
    - Draft technical documentation standards
5. **Jurisdictional Analysis**:
    - Document evidentiary standards across major jurisdictions
    - Compare procedural requirements for blockchain evidence
    - Analyse successful cross-jurisdictional cases
    - Identify common principles and jurisdiction-specific requirements
    - Develop an adaptable framework for a global application

6.2 Working Group Coordination

The BGIN IKP Working Group will coordinate these efforts through:

1. **Regular Meetings**: Monthly working group sessions to review progress and guide research
2. **Collaborative Workspaces**: Shared documentation and research repositories
3. **Expert Consultations**: Targeted discussions with subject matter specialists
4. **Public Workshops**: Open forums for broader community input
5. **Draft Circulation**: Regular sharing of work-in-progress for feedback
6. **Block Meeting Presentations**: Updates at BGIN Block gatherings

7. Call for Contributions

We invite contributions from:

1. **Analytics Companies**:
    - Insights on methodologies and challenges
    - Perspectives on balancing commercial interests with transparency
    - Experience with forensic applications of analytics tools
    - Understanding of validation procedures and error rates
    - Approaches to documentation and chain of custody
2. **Legal Experts**:
    - Perspectives on evidentiary standards across jurisdictions
    - Experience with blockchain evidence in court proceedings
    - Knowledge of expert witness qualification requirements
    - Understanding of chain of custody requirements
    - Interpretation of scientific evidence standards
3. **Law Enforcement**:
    - Practical experience with blockchain investigation
    - Challenges in presenting blockchain evidence
    - Training needs and resource limitations
    - Collaboration models with analytics companies
    - Case studies of successful prosecutions
4. **Academic Researchers**:
    - Technical and methodological insights
    - Peer review of proposed standards
    - Development of validation frameworks
    - Research on heuristic reliability
    - Analysis of error rates and statistical significance
5. **Regulatory Authorities**:
    - Guidance on compliance frameworks
    - Perspectives on standards development
    - Approaches to cross-border coordination
    - Balancing innovation with public protection
    - Implementation considerations for new standards
6. **Blockchain Companies**:
    - Technical perspectives on on-chain data
    - Insights on privacy-enhancing technologies
    - Understanding of transaction models and limitations
    - Knowledge of emerging blockchain architectures

- Perspective on the technical feasibility of forensic approaches

Conclusion / Discussion

The distinction between blockchain analytics and forensics represents a critical area for industry development. With proper standards and guidelines, blockchain forensics can mature into a robust, legally defensible discipline while analytics can continue to innovate in risk management and pattern detection.

The current state of blockchain analysis reveals significant tensions between proprietary commercial interests and the transparency demands of forensic evidence. The Bitcoin Fog case highlights the potential and limitations of current approaches, suggesting a need for more rigorous standards, particularly as blockchain technology expands globally.

BGIN's multi-stakeholder approach offers the ideal framework for developing these standards through collaborative effort and diverse expertise. By bringing together technical, legal, regulatory, and industry perspectives, we can create a framework that balances innovation with integrity, privacy with security, and commercial interests with public accountability.

The result of this work will be a comprehensive resource for all blockchain ecosystem participants, providing clarity on best practices, methodological standards, and evidentiary requirements. This will benefit law enforcement, courts, analytics companies, and the broader blockchain community by establishing consistent, transparent, and reliable standards for blockchain investigation.

We invite all interested parties to join us in this important work to shape the future of blockchain forensics and analytics.

Contact Information:

- Mitchell Travers - IKP WG Chair
- BGIN Secretariat: Amanda Wick

Appendix - Footnote / endnote : formatted for a journal based on the research and references completed to date > ongoing

BGIN Research Proposal

Block #11 Innovation and Challenges of Blockchain Analysis Workshop

Background

Recent discussions in the blockchain community have highlighted the need to distinguish between blockchain forensics and analytics, particularly regarding evidentiary standards and methodologies. The Bitcoin Fog case brought this distinction to the forefront, raising questions about what constitutes admissible blockchain evidence in court.

Research Objectives

1. Define clear distinctions between blockchain forensics and analytics
2. Establish methodological frameworks for each discipline
3. Develop standards for evidence admissibility in legal proceedings
4. Create guidelines for industry practitioners
5. *What is the ultimate goal of the paper?*
    a. Make the distinction between analytics (eye of the beholder) statistics on data
        i. The derivation of conclusions
    b. Forensics is the meticulous work of tracing and tracking tx's and establishing verifiable / confirmed connections between a person and a pubic key
        i. More of police / scientific work / to be used in expert testimony
6. Intra market specialisation between firms
    a. Analytics
    b. Investigations
    c. Evidence
7. is Analytics make claims, forensics confirms
    a. Conflict of interest if the same firm is used ?
8. Traditional cases where there is a conflict of interest between data and forensics
    a. Finding a case where there was a question between two expert witnesses re/data governance
9. Databases of connections between wallets
    a. These heuristics + verifiability of these is questionable, however useful in analytics but when required in forensics, should be *revealed*?
        i. Conflict between IP of blockchain analytics
            1. Evidence-based
10. Knowledge of writing and understanding blockchain data / inferences
11. Additional contributions from Beth*?
    a. Was providing the testimony in bitcoin fog
    b. Moti - (also interested)
12. Level set - between the academic / corporate / and the data companies
    a. Mastercard resulting - risk management exercise

        i. Ciphertrace crippled /
    b. Build a discussion around ‘whether the evidence is true’ or ‘untrue’
        i. This has impact / implications which are real
            1. REAL IMPACT THIS IS PEOPLES LIVES / JOBS
13. Find news/reports on the fallout of the post case
    a. Use as part of the context section.

Key Research Areas

Blockchain Forensics

- Evidentiary standards compliance (e.g., Daubert standard)
- Chain of custody requirements
- Documentation methodologies
- Expert witness qualification criteria
- Forensic tool validation requirements

Blockchain Analytics

- Data collection methodologies
- Statistical analysis frameworks
- Pattern recognition techniques
- Risk scoring models
- Clustering algorithms reliability

Proposed Methodology

1. Literature Review
    - Existing legal frameworks
    - Current industry practices
    - Academic research
    - Court precedents
2. Case Studies
    - Bitcoin Fog case analysis
    - Other relevant legal proceedings
    - Industry implementation examples
3. Stakeholder Interviews
    - Law enforcement agencies
    - Blockchain analytics companies
    - Legal experts
    - Academic researchers
4. Technical Analysis
    - Forensic tool evaluation
    - Analytics platform assessment
    - Evidence collection methods

    - Data verification procedures

Expected Outcomes

1. Framework Document
    - Clear definitions and distinctions
    - Methodological guidelines
    - Best practices recommendations
    - Implementation guidance
2. Standards Proposal
    - Evidence admissibility criteria
    - Tool validation requirements
    - Documentation standards
    - Quality assurance procedures
3. Industry Guidelines
    - Practitioner certification requirements
    - Tool evaluation criteria
    - Implementation recommendations
    - Risk management procedures

Proposed Research Team;

Editors

- Mitchell Travers - IKP WG Chair
- Amanda Wick - Steering Committee

Contributors

Suggestions:

- Advertise the research initiative
    - In the next two weeks for the presentation IKP wg call
    -
- Upload a summary of the research proposal
    - Social networks
    - Transparency for expert participation
    - BGIN Lead / individual contributors
        - Sensitive around company attributions
            - Analytics
-

BGIN's Multistakeholder Role

Unique Position

BGIN, as a multistakeholder discussion body and distributed think tank, is uniquely positioned to lead this research initiative due to:

- Diverse expertise across technical, legal, regulatory and industry domains
- Neutral stance as a non-commercial entity
- Global reach and perspective
- Track record in developing shared understanding of complex blockchain topics

Stakeholder Engagement

The research will actively involve key stakeholders including:

- **Analytics Companies**: Chainalysis, TRM Labs, Elliptic, Soliduis Labs and others to share methodologies and challenges
- **Law Enforcement**: Agencies using blockchain analytics/forensics tools
- **Exchanges**: Major crypto exchanges dealing with compliance requirements
- **Legal Experts**: Specialists in evidence standards and regulatory frameworks
- **Academic Researchers**: Contributing peer review and methodological rigor
- **Blockchain Companies**: Technical experts building underlying systems
- **Regulators**: Agencies developing compliance frameworks

Collaborative Development Process

1. **Working Group Formation**
    - Establish dedicated working group within BGIN IKP community
    - Regular meetings and workshops to drive progress
        i. Monthly IKP WG
2. **Community Input**
    - Public consultations on draft frameworks
    - Request for comments periods
    - Regional workshops to gather global perspectives
    - Integration of feedback from BGIN Block meetings
3. **Knowledge Sharing**
    - Development of educational resources
    - Training materials for practitioners
    - Case study documentation
    - Best practice guidelines
4. **Industry Alignment**
    - Coordination with existing standards bodies
    - Engagement with regulatory working groups
    - Collaboration with industry associations
    - Partnership with academic institutions

BGIN's Value Add

1. **Independent Perspective**
    - Non-commercial viewpoint
    - Focus on public good
    - Balanced consideration of competing interests
2. **Technical Credibility**
    - Deep blockchain expertise
    - Understanding of cryptographic principles
    - Experience with distributed systems
3. **Regulatory Insight**
    - Strong relationships with regulators
    - Understanding of compliance requirements
    - Global regulatory perspective
4. **Industry Trust**
    - Established reputation
    - History of successful initiatives
    - Broad industry participation

Implementation through BGIN

1. **Research Coordination**
    - BGIN secretariat to coordinate research activities
    - Working group leads to manage specific workstreams
    - Regular progress reporting to BGIN community
2. **Resource Development**
    - Creation of shared terminology
    - Development of technical standards
    - Documentation of best practices
    - Production of training materials
3. **Community Engagement**
    - Regular updates at BGIN Block meetings
    - Online collaboration platforms
    - Regional workshops and events
    - Public consultation periods
4. **Quality Assurance**
    - Peer review process
    - Technical validation
    - Legal review
    - Stakeholder feedback integration

Through this structured approach leveraging BGIN's unique position and multistakeholder model, the research can develop robust, widely-accepted standards for distinguishing blockchain forensics from analytics while ensuring practical applicability and broad adoption.

Resources

BGIN Block#11 MR Innovation and Challenges of Blockchain Analysis
https://bgin-global.org/
https://bgin.discourse.group/

Appendix

BGIN Block#11

Innovation and Challenges of Blockchain Analysis

Date: Monday, October 22, 2024

Summary

Introduction to Travel Rule Requirements: The Travel Rule in cryptocurrency introduces complexities, as digital assets like Bitcoin operate without identity requirements, making compliance challenging. The industry has developed out-of-band solutions to accommodate regulatory expectations without altering core blockchain protocols.

Industry Collaboration: In 2019, the crypto industry began working collaboratively to create a framework allowing regulated entities to share information to comply with the Travel Rule.

Challenges in Implementation: Implementing the Travel Rule is difficult due to the decentralized nature of blockchain technology, yet efforts persist to balance regulatory demands with decentralized principles.

Client Education and Risk Management: Proper configuration of risk rules and training for clients, especially those with regulatory or public sector responsibilities, are essential for interpreting blockchain analytics and handling suspicious activities effectively.

Global Regulatory Challenges: Addressing regulatory differences across jurisdictions requires a balance between local compliance and the global, borderless nature of blockchain technology. BGIN could support industry discussions by providing educational resources on these challenges.

Transparency and Competitive Concerns: Exchanges are pushing for more transparency in blockchain analytics, as current practices often lack clear evidence for high-risk scores. This “black-box” approach hinders client trust, and proprietary constraints limit open discussion.

Blockchain Analytics vs. Forensics: Distinguishing between blockchain analytics (data analysis) and forensics (evidence for prosecution) could clarify roles and improve accuracy. This distinction could benefit from an academic focus, especially regarding admissible evidence standards like the Daubert standard in U.S. law.

Emerging Proactive Detection Techniques: Proactive detection methods, including mempool analysis and opcode analysis, are evolving to preempt fraud by identifying risks before they manifest on-chain. This anticipatory capability is unique to blockchain and not found in traditional finance.

Data Privacy Concerns: The industry's rapid advancement in analytics raises concerns about data privacy and the risk of creating a pervasive surveillance system. There is a call for more nuanced discussions around data privacy before building potentially irreversible surveillance structures.

AML and KYC in Blockchain Context: Anti-money laundering (AML) and Know Your Customer (KYC) standards in crypto compliance involve transaction monitoring and are evolving to include broader blockchain analytics and proactive fraud detection.

Philosophical and Ethical Implications: There is a debate over consumer protection in a decentralized financial environment, where some clients resist the idea of censoring transactions, believing that the responsibility lies with the users.

Staking and Block Building Responsibilities: Institutions participating in staking and block-building must consider compliance with sanctions and ethical responsibilities, such as avoiding risky or sanctioned transactions. This responsibility extends to developing compliant, competitive blocks in decentralized finance.

Emerging Market for Ethical Block Builders: As Web3 matures, there is potential for a new market of ethical block builders, akin to the sustainable investing trend. These entities could prioritize environmental and ethical standards, such as renewable energy usage and avoidance of nefarious activities.

Importance of Reliable Methodologies: BGIN could contribute by promoting accurate clustering and analytics methodologies to reduce false positives/negatives, as inaccuracies can significantly impact individuals. Case study materials may help establish a common ground for analytics providers and users.

BGIN Research: Forensics vs Analytics in Blockchain Analysis

Next Actions:

+ Share relevant research/insights in chat - focus on evidence standards & methodology distinctions
+ Deep dive into Bitcoin Fog case implications needed
+ Looking for TradFi precedents where data evidence validity was challenged
+ Need subject matter experts in forensics/analytics - introductions welcome
+ Group sync mid/late Jan (pre-IKP WG Jan 30)
+ Block 12 deliverable: Draft framework + expert workshop

Context:
Recent cases exposed critical gaps between blockchain forensics and analytics methodologies. Key question: When does pattern analysis become admissible evidence? Analytics makes probabilistic claims based on patterns. Forensics requires verifiable connections between individuals and keys. This distinction has significant implications for legal proceedings and investigations.

The research examines:
*Evidence standards in digital forensics
*Chain of custody in blockchain analysis
*Conflicts between proprietary tools and transparency
*Impact on criminal proceedings

Particularly interesting is the tension between commercial blockchain analysis tools and legal evidence requirements. Several cases highlight where this distinction became critical.

Current phase: Building framework through case studies and expert input. Block 12 workshop will focus on practical implementation.
Drop research links/thoughts here 🔍

16th Jan:

References:

Bitcoin Fog Case Summary.pptx
Bitcoin Fog Case Daubert Hearing Summary
BGIN Block#11 MR Innovation and Challenges of Blockchain Analysis

Agenda

1. Review research on bitcoin fog from Takaya
2. Complete next steps and focus on the paper
3. Prepare for the presentation to IKP WG (draft) on the 30th
4. Delegate
    a. Call to contribution
5. Advertising
    a. Code
    b. Strategy
6. Paper editors and contributors.
    a. Next steps / tasks

Timeline:Jan - March

1. First sync Jan 16
    a. Delivering research Bitcoin Fog / Takaya
2. First draft presentation of research
    a. IKP WG
    b.
3. Preparation of draft paper
4. Advertising and promotion of the paper efforts
5. Begin conducting interviews with experts
6. Deliver any require research
7. BGIN Block 12
    a. Workshop 90mins with all contributors
    b. Next steps discussion
8. Paper Continue editing

Notes:

- Academic peer reviews
- Forensic procedure
    - Have to be peer review'able
- In order for something to be considered evidence, something should be peer reviwable
- Difference between heuristics

        - Heuristic techniques
            - Black box
            - Data set
        - Possibility for conflicting manipulation of the data
- Overlap between analysis companies - peer review
    - Transition from analysis to forensics
- The way heuristic is used here:
    - Replace is ‘rule of thumb’
        - Same thing?
    - Arbitrary
- Our understanding of facts are based on heuristics which are arbitrary
- Contentious
    - **Analysis of the word heuristic**
- Which are we reviewing
    - Techniques - peer review?
    - Results - peer review?
        - BOTH?
-

Data Analysis vs Forensic Analysis

- Each clustering technique combines multiple data types depending on the purpose or case.
- The use of techniques tailored to non-universal human behaviors, product characteristics, or specific cases, especially when utilizing data not publicly accessible, can be defined as forensic analysis.

Clustering Techniques	Data Type (1) on-chain data	Data Type (2) (2) Data identified through data leaks and undercover investigation	Data Type (3) (3) Data identified through governmental subpoena power	Data Type (4) (4) Data shared by entities
Clustering based on technical and scientific theories (e.g., co-spend clustering, multi-input clustering)	This is the primary data type used. Transaction histories and relationships between addresses are analyzed.	May be used as supplemental information (e.g., for verification or to provide additional context for clusters)	Typically not used, but legally obtained data might contribute to improving clustering accuracy in certain cases	Can be used when provided
Clustering based on the behavioral patterns of individuals or organizations and the characteristics of products.	Used to analyze how specific groups of addresses are being analyzed	May help identify behavioral patterns	Can be used to supplement details about behavioral patterns	Used when behavioral data is directly provided by clients or services
Clustering based on manual processes tailored to specific cases using data such as attribution	May not always be the primary data type but can be used depending on the case	Critical in specific case-based manual processes	Used to support manual approaches in specific cases	Frequently used for custom solutions tailored to specific cases

source: https://storage.courtlistener.com/recap/gov.uscourts.dcd.232431/gov.uscourts.dcd.232431.159.1.pdf
https://caselaw.findlaw.com/court/us-dis-crt-dis-col/115886148.html

Data analysis vs Forensic analysis

- A mismatch between the providers of sources of.

Technical Resources on the Discussion
- Whether or not things are peer-reviewed?
- For a Daubert analysis versus academic peer review
    - Generally publicly available
    - Published the results in a paper.

Heuristics section - crucial aspect of the problem + analysis
The list of heuristics - find the sources
Peer review of the logic of the Heuristic

It is more complicated for the company to implement these.

How does the process around blockchain analysis or forensics
/
Investigator analysis is an opinion.

When the tools claim to be fact

The line between whats an investigator and what is a tool response

What is the verification process / the company.

The best practice element.

Heuristics are a risky path.

How are they relevant, what should the burden of proof be.

It's a combination of legal meets product.

- Heuristics are arbitrary and can be chosen to confirm analysis
- Non-definitive
- Contentious
    - Attribution side
    - Probability discussion
    - Disclosure discussion

Clustering

- Issues
    - False positives
    - Big clustering

Data sourcing
Attributions

If the process is the same, the results should be the same
Regardless of the deployer
Reproducing the same process. - as a test.

The thesis:

Forensics vs Analysis is the discussion and open question in the paper

There is value in explaining the difference - building on the standards and clarifying the difference between and aspect of a tool versus an investigator. Methods
Based on my knowledge and experience
Verses
What is the tool being used for from a forensic standpoint?
Daubert is an analysis of the tool.

Investigations are a craft.

Patterns are programmatic or human-derived

The primary mechanism of tools is to contextualise the blockchain—user context improvement experience to the data layer of understanding.

Difference between

Investigator paths throughout the data are the craft

The tool is how the data is presented.
Does this need to be understood as a difference?

Data about where you can then request off-chain data.

Courts will see the value in this conversation as an education and prep material for sharing to protect the

Explaining via BGIN the role in each of the steps and where the tool is impacting and implementing their role in the investigation and case.

Risks need to be managed.
Purpose of explaining the forensic vs analysis
Best practices - what is it okay to rely on the tool for
What skill set does an investigator need to be valid?

What is a good expert witness for analytics?

Parallel reconstruction
Show your work by hand.

Identify the difference between what is proprietary about a heuristic and what is peer-reviewed (reviewable).

What is appropriate for an investigator, and what is appropriate for a tool?

Difference between USA filing using Chainalysis report and Canada
- Please provide reference for this (this is key)
    - 3rd party reports need to be reconstructed.

BGINs role in speaking about the differences. Presenting whats on the table, the questions the useful higher level of education - path to the discussion.

Where the tool ends and where the investigator's work begins.

Meeting Report: BGIN Research on Blockchain Forensics vs Analytics Date: January 16, 2025

Key Discussion Points:

Research Review

- Reviewed Takaya's analysis of the Bitcoin Fog case and its implications for blockchain forensics standards
- Examined the distinction between forensics and analytics in blockchain investigation
- Discussed the role of heuristics in blockchain analysis tools and their admissibility as evidence

Heuristics Analysis

- Explored the complexity of heuristic techniques in blockchain analysis
- Identified concerns about the "black box" nature of proprietary analysis tools
- Discussed the need for peer review standards in both techniques and results
- Noted the contentious nature of heuristic-based analysis and its limitations

Forensics vs Analytics Framework

- Examined the distinction between investigator analysis (opinion-based) and tool-generated results
- Discussed verification processes and best practices
- Identified the need to clarify where tool functionality ends and investigator work begins
- Explored the relationship between programmatic patterns and human-derived analysis

Legal Considerations

- Discussed differences between US and Canadian approaches to third-party blockchain analysis reports
- Examined Daubert standard requirements for tool validation
- Highlighted the importance of parallel reconstruction in validating tool results
- Noted the need for clear standards in expert witness qualification

BGIN's Role

- Identified opportunity to provide educational resources and framework development
- Discussed potential for establishing best practices in blockchain investigation
- Emphasized importance of maintaining balanced, multi-stakeholder perspective

Next Steps:

1. Prepare draft presentation for IKP WG meeting on January 30th
2. Begin outreach for expert interviews and contributions
3. Develop advertising strategy for research initiative

4. Continue drafting paper framework
5. Plan workshop for BGIN Block 12

Timeline (January-March):

- January 30: IKP WG presentation
- February: Expert interviews and research continuation
- March: BGIN Block 12 workshop and paper development

Action Items:

- Collect references on US/Canadian filing differences
- Develop framework for distinguishing proprietary vs peer-reviewable heuristics
- Create outline for best practices documentation
- Begin drafting educational materials on forensics vs analytics distinction
- Find developer resources for the more engineering focused side of this paper.
    - Academic
    - Former Devs from analysis companies

The meeting emphasized the importance of establishing clear standards for blockchain investigation while recognizing the distinct roles of automated tools and human investigators. BGIN's position as a neutral forum for these discussions was highlighted as crucial for developing balanced, practical guidelines for the industry.

1. Forensics v Analysis
    a. Brief
    b. Updated work items
        i. BGIN IKP Working Group Distinguishing Blockchain Forensics fr…
        ii. Distinguishing Blockchain Forensics from Analytics2025jan30_w…
            1. See

BGIN IKP Working Group Meeting Notes

Date: January 30, 2025 Topic: Blockchain Forensics vs Analytics Research Paper

Key Discussion Points

Cross-Jurisdictional Considerations

- Current case studies are US-centric (e.g., Bitcoin Fog case) with specific legal standards
- Need to expand analysis to include other jurisdictions' approaches to blockchain analysis
- Identified Canadian cases for potential inclusion:
    - Dark web narcotrafficking case
    - $1.4M Bitcoin wallet seizure case
- Need input from law enforcement officials across jurisdictions regarding:
    - Methodological differences
    - Resource and tool access variations
    - Capacity constraints

Standards and Guidelines

- Goal to develop cross-jurisdictional standardization framework
- BGIN identified as potential facilitator for global jurisdictional analysis
- Need to engage crown prosecutors and similar authorities in different regions

Technical Analysis Framework

1. Forensic Techniques vs Attribution Data
    - Attribution data identified as proprietary technology
    - Importance of combining on-chain and off-chain data
    - Need for proper contextualization
2. Data Science Fundamentals
    - Need to establish "blockchain science" fundamentals
    - Industry standards discussion (e.g., CoinJoin analysis)
    - Service/product categorization from organizations

Interview Strategy Refinement

Current challenges:

- Questions may be too broad (e.g., "what is forensic vs analysis")

- Need for better context setting

Proposed improvements:

- Develop educational material for interviewees
- Create fact file for concrete discussion points
- [Action Item] Mitchell to produce contextual information package

Quality Assurance Framework

Need to establish standards for:

- Technical (data science) methodology
- Practitioner qualifications
- Data quality and verification

Next Steps

1. Create brief (one-page) technical resource document covering:
    - Core chain-based fundamentals
    - High-level concepts
    - Key definitions
2. Develop interview framework that will:
    - Provide clarity on core concepts
    - Address fact vs inference derivation
    - Cover behavioral analysis methodology
3. Expand jurisdictional research beyond US cases

Action Items

- Develop educational materials for interviewees
- Create fact file for structured discussions
- Research additional jurisdictional case studies
- Draft technical resource document

Block 13

Elevating the general knowledge and understanding of analytics for judges / expert witness training.

Analogies are powerful for shared understanding - style of writing

- Are there any relatable documents in tradfi
- The example of smurfing *$9999 tx
    - Analysis of activity, and conclusions drawn from the typology
-

Standard document or reference.

Informed Opinion vs method based outcomes

Can this document be used as a general document

Baseline level information:

Standardisation

- Heuristics
    - Eg co-spend analysis
- Methodologies

Techniques which are mentioned in scientific papers (therefore publically referenceable)

Are the starting point

Data hygiene*

- Sourced attributions
- Not the investigator's responsibility
- Could be an issue
-

Equipt both prosecution and defence with general / level'd standard of knowledge

Objective resource document

Level the playing field - without requiring the service from experts in the tools, investigation and/or evidence

Resource intensity -

Clarify the aspects which do not rely on analytics to

Raw blockchain data replication (w/o tool subscription)

The sophistication of ML on-chain is only increasing, this game isnt fair for parallel reconstruction

parallel reconstruction - UK/Canada is more required (common practice |best/suggested practice)

The value of it is significant - explanation of this to agents/prosecutors/defence

High level conceptual / what tools do and can help with.

Find the public papers which share these things.

At what stage does the tool company need to come in ?

Investigator > Tool >

What questions are okay?

Not what are the answers to the questions

Resource guide.

Daubert - where it would apply, when it is appropriate to have the hearing?

- Section

Next steps:

Outline

‘Resource guide structure’

Represenative of multi jurisdictions

- UK, Japan, Canada